

CHOKURDAKH, Russian Federation

WMO: 219460

Lat: 70.617N

Lon: 147.883E

Elev: 200

StdP: 14.59

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -48.1	(c) -44.5	(d) -52.3	(e) 0.3	(f) -47.7	(g) -48.9	(h) 0.3	(i) -44.2	(j) 26.3	(k) -11.9	(l) 22.6	(m) -15.4	(n) 5.7	(o) 230	(p) 0.984

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.3	(c) 74.5	(d) 60.7	(e) 69.0	(f) 58.4	(g) 64.2	(h) 55.3	(i) 62.0	(j) 72.2	(k) 59.0	(l) 67.5	(m) 55.9	(n) 63.7	(o) 8.7	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.3	70.6	66.5	54.2	63.0	64.5	51.1	56.1	60.4	27.9	72.2	25.8	67.5	23.8	64.0	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.5	(b) 20.6	(c) 17.9	(e) -50.8	(f) 80.7	(g) 3.3	(h) 3.5	(i) -53.1	(j) 83.2	(k) -55.1	(l) 85.2	(m) -56.9	(n) 87.2	(o) -59.3	(p) 89.7		
(4) 24.5	(5) 20.6	(6) 17.9	(7) -50.9	(8) 65.7	(9) 3.2	(10) 2.6	(11) -53.3	(12) 67.6	(13) -55.2	(14) 69.1	(15) -57.0	(16) 70.5	(17) -59.3	(18) 72.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-27.5	-27.3	-16.1	1.6	23.0	43.9	50.4	46.7	35.0	13.7	-9.4	-23.5		
	DBStd	30.53	9.64	9.97	11.72	11.51	10.16	9.88	8.88	7.73	6.64	10.70	11.70	11.09		
	HDD50	15038	2402	2165	2048	1451	837	237	105	159	450	1126	1781	2277		
	HDD65	20296	2867	2585	2513	1901	1301	637	462	567	899	1591	2231	2742		
	CDD50	233	0	0	0	0	1	53	119	59	1	0	0	0		
	CDD65	16	0	0	0	0	0	4	11	1	0	0	0	0		
	CDH74	134	0	0	0	0	0	41	85	8	0	0	0	0		
	CDH80	13	0	0	0	0	0	3	10	0	0	0	0	0		
	WSAvg	8.5	7.7	7.7	7.7	8.0	9.3	10.1	9.8	9.0	8.5	8.0	8.3	8.0		
	PrecAvg	8.4	0.5	0.5	0.4	0.3	0.4	0.8	1.3	1.0	0.9	0.9	0.7	0.5		
	PrecMax	14.1	1.3	1.2	1.1	1.0	1.1	2.6	4.6	2.9	2.6	2.5	2.2	1.1		
(15) Precipitation	PrecMin	3.6	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.4	0.2	0.1		
	PrecStd	2.2	0.3	0.3	0.2	0.2	0.2	0.6	0.9	0.7	0.5	0.4	0.4	0.3		
	DB	2.4	1.6	19.6	32.5	56.0	79.4	80.8	75.6	59.1	33.3	21.6	10.6			
	MCWB	2.0	1.1	18.5	28.5	46.9	59.2	65.4	62.9	50.1	31.7	21.4	9.9			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	-3.7	-4.2	12.7	27.5	46.4	72.7	76.1	69.7	52.5	31.4	16.5	2.5			
	2% MCWB	-4.1	-4.6	11.5	24.8	40.1	57.1	62.5	58.3	46.7	30.6	16.0	1.8			
	5% DB	-8.4	-8.6	6.5	22.8	40.7	65.6	70.9	64.7	48.0	29.6	11.4	-2.2			
	5% MCWB	-8.9	-9.0	5.3	20.8	36.3	54.4	60.4	55.8	43.7	28.7	10.8	-2.7			
	10% DB	-13.1	-12.4	0.8	18.2	36.8	60.1	65.7	60.0	43.9	27.2	6.8	-7.2			
	10% MCWB	-13.3	-12.8	0.1	16.6	33.6	51.1	57.4	53.2	41.1	26.2	6.2	-7.6			
	0.4% WB	1.9	1.1	18.6	28.8	47.0	61.3	66.6	63.5	51.5	32.3	21.6	10.0			
	0.4% MCDB	2.2	1.4	19.5	31.8	55.9	75.4	78.1	73.5	56.8	32.8	21.7	10.6			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-4.1	-4.8	11.6	24.8	40.3	58.0	63.6	59.7	47.3	30.8	16.0	2.0			
	2% MCDB	-3.7	-4.5	12.8	27.5	45.8	70.4	74.1	67.6	51.8	31.4	16.6	2.5			
	5% WB	-8.8	-9.0	5.4	20.8	36.5	54.6	60.9	56.4	43.9	28.8	10.9	-2.7			
	5% MCDB	-8.4	-8.7	6.4	22.7	40.3	64.9	69.9	63.2	47.4	29.5	11.4	-2.2			
	10% WB	-13.4	-12.8	0.1	16.7	33.8	51.3	57.9	53.6	41.2	26.1	6.3	-7.6			
	10% MCDB	-13.1	-12.4	0.8	18.2	36.6	59.9	65.3	59.7	43.6	26.9	6.8	-7.3			
	MDBR	10.8	10.4	12.1	14.2	11.1	14.6	14.3	12.8	8.1	8.3	10.5	10.7			
	5% DB	16.2	12.5	15.5	17.0	14.2	22.7	21.6	20.8	15.6	6.6	13.1	16.1			
(36) Mean Daily Temperature Range	5% MCWBR	16.0	12.3	14.6	15.4	9.7	12.4	12.3	11.5	10.2	6.1	13.0	16.0			
	5% WB	16.1	12.6	15.4	16.6	13.8	22.2	20.3	18.7	14.5	6.6	13.1	16.0			
	5% MCWBR	15.9	12.4	14.6	15.1	9.6	12.9	12.5	11.5	10.0	6.0	13.1	15.9			
	Clear-Sky Solar Irradiance	taub	N/A	0.205	0.278	0.318	0.358	0.351	0.395	0.360	0.306	0.227	N/A	N/A		
(40) All-Sky Solar Radiation	taud	N/A	2.013	2.074	2.055	2.137	2.427	2.303	2.383	2.431	2.069	N/A	N/A			
	Ebn at Noon	0	161	222	251	256	266	245	241	223	141	0	0			
	Edh at Noon	0	12	25	36	38	30	33	27	18	10	0	0			
(44) All-Sky Solar Radiation	RadAvg	0	95	607	1375	1887	1766	1491	1020	475	146	0	0			
	RadStd	0	10	31	49	117	169	115	128	40	16	0	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+2.21	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page